

1 Stochastics and Dynamics
 2 Vol. 18, No. 6 (2018) 1950015 (28 pages)
 3 © World Scientific Publishing Company
 4 DOI: 10.1142/S0219493719500151



5 Asymptotic analysis and homogenization of invariant measures

6 Oleksandr Misiats
 7 *Department of Mathematics, Virginia Commonwealth University,*
 8 *Richmond, VA, 23284, USA*
 9 *omisiats@vcu.edu*

10 Oleksandr Stanzhytskyi
 11 *Department of Mathematics,*
 12 *Kyiv National University, Kyiv, Ukraine*
 13 *ostanzh@gmail.com*

14 Nung Kwan Yip
 15 *Department of Mathematics, Purdue University,*
 16 *West Lafayette, IN, 47907, USA*
 17 *yip@math.purdue.edu*

18 Received 24 October 2016
 19 Revised 17 February 2018
 20 Accepted 17 February 2018
 21 Published

22 In this paper, we study limiting behavior of the invariant measures for reaction–diffusion
 23 equations in the whole space $\mathbb{R}^d$ with regular and singular perturbations. In the regular
 24 case, we show the convergence of the unique stationary solution of $du_\varepsilon = (\Delta u_\varepsilon - \varepsilon u)dt +$
 25 $\sigma(x, u_\varepsilon)dW$ to a stationary solution of the limiting equation $du = \Delta udt + \sigma(x, u)dW$. We
 26 also consider the asymptotic behavior of the stationary solution under the perturbations
 27 of spectrum. Finally, for the singular perturbation of homogenization type, we show the
 28 weak convergence of invariant measure to its homogenized limit.

29 *Keywords:* Vanishing dissipation limit; stationary solution; long time homogenization
 30 limit.

31 *AMS Subject Classification:* 60G57, 60H15, 35B27

32 1. Introduction and Main Results

33 The theory of reaction–diffusion equations combined the theories of heat conduc-
 34 tion and mass diffusion, on the one hand, and equations of chemical and biological
 35 kinetics, on the other. A comprehensive study of the deterministic reaction–diffusion
 36 equation was performed by Volpert [40]. In this book, one can find numerous exam-
 37 ples of application of reaction–diffusion models.

O. Misiats, O. Stanzhytskyi & N. K. Yip

1 An important class of reaction–diffusion equations describes the dynamics in
2 random media, which leads to stochastic PDEs

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) = Du(t, x) + f(x, u(t, x)) + \sigma(x, u(t, x)) \dot{W}(t, x), & t > 0, \ x \in G \subset \mathbb{R}^d; \\ u(0, x) = u_0(x). \end{cases} \quad (1.1)$$

3 Here, D is an elliptic operator, f and σ are measurable functions, the Gaussian noise
4 $\dot{W}(t, x)$ is white in time and colored in space (see (1.2) for the precise definition
5 of $W(t, x)$), and $G \subseteq \mathbb{R}^d$. A comprehensive analysis of stochastic equations of this
6 type has been performed by Da Prato and Zabczyk (see [13, 15] and references
7 therein). The long time behavior of solutions of (1.1) is a question of a special
8 interest. In particular, long time behavior of solutions of (1.1) is closely related
9 to the existence of invariant measures. The works [13, 15] provide conditions for
10 the existence and uniqueness of invariant measures for (1.1) based on the results of
11 Krylov and Bogoliubov [24] on the tightness of a family of measures. The conditions
12 for the existence of invariant measures were expressed in terms of the coefficients
13 in [28, 21, 6, 8–10] in particular, in bounded domains. An alternative approach
14 to establishing the existence of invariant measures, which is based on the coupling
15 method, was suggested by Bogachev and Rockner [5] and Mueller [33]. This method
16 can be applied even for space-white noise but only in the case when the space
17 dimension is one.

18 The solutions of (1.1) in unbounded domains are studied in weighted Sobolev
19 spaces. Let ρ be a non-negative continuous $L^1(\mathbb{R}^d) \cap L^\infty(\mathbb{R}^d)$ function. Following
20 [38], we call ρ to be an *admissible* weight if for every $T > 0$ there exists $C(T) > 0$
21 such that

$$G(t, \cdot) * \rho \leq C(T)\rho, \quad \forall t \in [0, T], \quad \text{where } G(t, x) = \frac{1}{(4\pi t)^{d/2}} e^{-\frac{|x|^2}{4t}}.$$

22 Typically, we set $\rho(x) = \exp(-\gamma|x|)$ for $\gamma > 0$, or $\rho(x) = (1 + |x|^n)^{-1}$ with $n > d$,
23 which are admissible weights. For such weights, we define

$$L_\rho^2(\mathbb{R}^d) := \left\{ w : \mathbb{R}^d \rightarrow \mathbb{R}, \int_{\mathbb{R}^d} |w(x)|^2 \rho(x) dx < \infty \right\}.$$

24 Throughout the paper, we will use the notation

$$\|w\|_\rho^2 := \int_{\mathbb{R}^d} |w(x)|^2 \rho(x) dx$$

25 to denote the weighted L^2 norm in $\mathbb{R}^d$. The $L^2(\mathbb{R}^d)$ norm without weights will be
26 denoted with $\|\cdot\|$. The noise W is given by

$$W(t, x) := \sum_{k=1}^{\infty} \sqrt{a_k} e_k(x) \beta_k(t), \quad (1.2)$$

1 where $\beta_k(t)$ are independent Wiener processes, $\{e_k(x), k \geq 1\}$ is an orthonormal
 2 basis in $L^2(\mathbb{R}^d)$ such that

$$\sup_{k \geq 1} \|e_k\|_{L^\infty(\mathbb{R}^d)} \leq 1$$

3 and $a_k \geq 0$ satisfy

$$\sum_{k=1}^{\infty} a_k := a < \infty. \quad (1.3)$$

4 Clearly, in a bounded domain G an elliptic operator with homogeneous Dirichlet
 5 boundary conditions has discrete negative spectrum, separated from the origin.
 6 Thus

$$\|S(t)u\|_{L^2(G)} \leq Ce^{-\lambda_1 t} \|u\|_{L^2(G)}, \quad u \in L^2(G), \quad (1.4)$$

7 where $\lambda_1 > 0$ is the first eigenvalue of $-D$ in G (subject to Dirichlet boundary
 8 condition $u = 0$ on ∂G), and $S(t)$ is the corresponding semigroup generated by
 9 D . Using (1.4) one can easily establish the existence and uniqueness of invariant
 10 measures [31] even in certain unbounded domains.

11 In unbounded domains, however, the spectrum is typically more complicated,
 12 thus the validity of the estimate of type (1.4) is a highly nontrivial question. A
 13 typical approach to establish the existence of invariant measures in unbounded
 14 domains is to require some form of dissipation from the nonlinearity f . An example
 15 of such dissipation condition can be found in [2]:

$$uf(u) \leq -ku^2 + c \quad (1.5)$$

16 for some $k > 0$ and $c \in \mathbb{R}$. The work [31] established the existence of invariant
 17 measure for (1.1) with $D = \Delta$ in $\mathbb{R}^d$ under different conditions on f . In particular,
 18 the invariant measure for (1.1) exists if f satisfies the global bound:

$$|f(x, u)| \leq \varphi(x) \in L^1(\mathbb{R}^d) \cap L^\infty(\mathbb{R}^d), \quad \forall u \in \mathbb{R}.$$

19 In [32], using a version of Ito's formula, the dissipation condition (1.5) is replaced
 20 with a weaker one:

$$uf(x, u) \leq \eta(x) \quad \text{for } |u| > M \text{ and } x \in \mathbb{R}^d$$

21 for some $M > 0$ and a non-negative $\eta(x) \in L^1(\mathbb{R}^d)$, $d \geq 3$. The question of the
 22 existence of invariant measures in unbounded domains was also addressed in [15,
 23 19, 38].

24 The uniqueness of invariant measure is a question of a separate interest [15]. In
 25 general, to show uniqueness, one needs the recurrence property for the solutions, as
 26 well as the strong Feller property. However, if the semigroup satisfies an estimate
 27 of type (1.4), the uniqueness of the invariant measure follows easily, e.g. using the
 28 method developed in [31]. More precisely, we have the following theorem for the
 29 Eq. (1.1) with solution u taking values in a Hilbert space H .

O. Misiats, O. Stanzhytskyi & N. K. Yip

Theorem 1.1. ([31]) *Let $S(t)$ be the semigroup for the operator D such that*

$$\|S(t)u\|_H \leq Ce^{-\lambda t}\|u\|_H, \quad \forall u \in H, \quad \lambda > 0. \quad (1.6)$$

Then, if $f(x, u)$ and $\sigma(x, u)$ have sufficiently small Lipschitz constants in u , there exists a unique stationary solution $u^ \in H$ for (1.1). This stationary solution defines the unique invariant measure for (1.1) in H . Moreover, this solution can be constructed as a limit of an iteration scheme.*

Remark 1.1. The semigroup, generated by $\Delta u - \varepsilon u$ in $H = L^2(\mathbb{R}^d)$ satisfies (1.6). The other examples of semigroups satisfying (1.6) are provided in [31].

However, there is a large class of practically important operators, e.g. the Laplace operator in $\mathbb{R}^d$, that do not satisfy (1.6) yet for which (1.1) has a stationary solution. In contrast with the previous theorem, the existence results for stationary solutions are rather abstract and not constructive ([2]). Moreover, we cannot say anything about the uniqueness of the stationary solution in this case. Nevertheless, in certain cases we can approximate the original operator D with the family of operators D_ε , which satisfy (1.6) with $\lambda = \lambda(\varepsilon) \rightarrow 0, \varepsilon \rightarrow 0$. This way we can efficiently approximate one of the stationary solutions of Eq. (1.1). We have the following results:

Theorem 1.2. (Vanishing dissipation limit) *Consider*

$$u_t^\varepsilon = \Delta u^\varepsilon - \varepsilon u^\varepsilon + \sigma(x, u^\varepsilon)\dot{W}(t) \quad (1.7)$$

and the limiting equation

$$u_t^* = \Delta u^* + \sigma(x, u^*)\dot{W}(t). \quad (1.8)$$

Assume $\sigma(x, 0) \in L^2(\mathbb{R}^d)$ with $d \geq 3$, and σ is Lipschitz in u with a sufficiently small Lipschitz constant L satisfying

$$2a \left(1 + \frac{1}{(8\pi)^{d/2} \left(\frac{d}{2} - 1 \right)} \right) L^2 < 1. \quad (1.9)$$

Let u_ε^ be the unique stationary solution of (1.7) (whose existence follows from Theorem 1.1). Then there exists u_0^* , which is a stationary solution of (1.8), such that*

$$E\|u_\varepsilon^*(t) - u_0^*(t)\|_{L^2(\mathbb{R}^d)}^2 \rightarrow 0, \quad \varepsilon \rightarrow 0, \quad t \geq 0.$$

Note that while the limiting equation may have more than one stationary solution, the previous theorem enables us to approximate one of them as a vanishing dissipation limit. It will be interesting to investigate in the future more precisely which limit is to be selected. A closely related result on the convergence of invariant measures to the invariant measure of the limiting equation was obtained in [29] in the finite-dimensional case.

In the next theorem, we study the behavior of the invariant measures under the regular perturbations of differential operators in $\mathbb{R}^d$ with discrete spectrum. We will make use of the results from the spectral theory.

Theorem 1.3. ([39, Chap. 19]) *Consider the spectral problem*

$$\Delta u - q(x)u + \lambda u = 0, \quad u \in L^2(\mathbb{R}^d), \quad \lambda \in \mathbb{R} \quad (1.10)$$

and the corresponding perturbed problem

$$\Delta u_\varepsilon - q(x)u_\varepsilon - \varepsilon q_1(x)u_\varepsilon + \lambda_\varepsilon u_\varepsilon = 0, \quad u_\varepsilon \in L^2(\mathbb{R}^d), \quad \lambda_\varepsilon \in \mathbb{R}. \quad (1.11)$$

Suppose q and q_1 are such that

- (i) q and q_1 are continuous;
- (ii) $q(x) \geq 0$, $q(x) \rightarrow +\infty$ as $|x| \rightarrow \infty$.

Then (1.10) has discrete spectrum $\{\lambda_n, n \geq 1\}$, with $\lambda_1 < 0$. The corresponding eigenfunctions are denoted with $\{\Psi_n\}$. If, in addition,

- (iii) *there exists $M > 0$ such that $q_1(x) \geq -Mq(x)$;*
- (iv) *for all $n \geq 1$ we have $q_1(x)\Psi_n(x) \in L^2(\mathbb{R}^d)$, then (1.11) has discrete spectrum $\{\lambda_n^\varepsilon, n \geq 1\}$ as well, with the corresponding eigenfunctions Ψ_n^ε . Moreover, for every $n \geq 1$ we have*

$$\lambda_n^\varepsilon = \lambda_n + O(\varepsilon), \quad \varepsilon \rightarrow 0$$

and

$$\int_{\mathbb{R}^d} (\Psi_n^\varepsilon(x) - \Psi_n(x))^2 dx = O(\varepsilon^2), \quad \varepsilon \rightarrow 0.$$

We establish the next theorem.

Theorem 1.4. (Perturbation analysis) *Consider*

$$du = [\Delta u - q(x)u - \varepsilon q_1(x)u + f(x, u)]dt + \sigma(x, u)dW(t) \quad (1.12)$$

and

$$du = [\Delta u - q(x)u + f(x, u)]dt + \sigma(x, u)dW(t). \quad (1.13)$$

Assume that q and q_1 satisfy the conditions of Theorem 1.3, and $f(x, u)$ and $\sigma(x, u)$ are continuous in x and Lipschitz in u , with $f(x, 0) \in L^2(\mathbb{R}^d)$, $\sigma(x, 0) \in L^2(\mathbb{R}^d)$, and the Lipschitz constant L satisfying

$$L^2 \left(\frac{16}{\lambda_1^2} - \frac{4a}{\lambda_1} \right) < 1 \quad (1.14)$$

and

$$\frac{\lambda_1}{2} - 24(1+a)\frac{L^2}{\lambda_1} < 0. \quad (1.15)$$

Here, $\lambda_1 < 0$ is the first eigenvalue of (1.10), and a is given by (1.3). Let u_ε and u^ be the unique stationary solutions of (1.12) and (1.13), respectively. Then, for every $t \geq 0$ we have*

$$\mathbb{E} \|u_\varepsilon(t) - u^*(t)\|_{L^2(\mathbb{R}^d)}^2 \rightarrow 0, \quad \varepsilon \rightarrow 0.$$

O. Misiats, O. Stanzhytskyi & N. K. Yip

1 **Example 1.1.** Suitable q and q_1 in $\mathbb{R}^2$ are $q(x, y) = x^2 + y^2$ and $q_1(x, y) = xy$.

2 A separate class of problems involves the study of porous media, subject to
3 the influence of random noise. This leads us to the homogenization problems for
4 stochastic differential equations. A comprehensive study of linear homogenization
5 problems in both elliptic and parabolic settings, as well as certain homogenization
6 results for elliptic operators in random media, may be found in [43]. In particular,
7 the work [43] addresses the homogenization problem in presence of regular ran-
8 dom perturbations. In this case, it is possible to treat the corresponding stochastic
9 equations for almost every realization of the random field, and obtain the results
10 in spirit of Central Limit Theorem. This approach however cannot be extended to
11 the singularly perturbed stochastic equation with white noise.

12 One of the earlier works, which studies the nonlinear homogenization problem
13 for parabolic stochastic equations in bounded domains is [3]. The consequent work
14 [23] considers the same problem with an oscillating coefficient in the noise term. The
15 long time behavior of stochastic homogenization problems was considered in [42]
16 in the linear case in a bounded domain. The work [4] studies the homogenization
17 problem for Brinkman flows. In this case, the oscillations are present in the reaction
18 terms, and the authors establish the convergence in probability.

19 In this paper, we characterize the convergence of invariant measures in the
20 homogenization limit. More precisely, we establish the mean square convergence
21 of the unique stationary solution of the nonlinear stochastic equation with rapidly
22 oscillating coefficients to the corresponding unique stationary solution of the homog-
23 enized problem. This, in turn, implies the weak convergence of invariant measures.
24 The two-scale convergence [1] and Nash–Aronson estimates play an important role
25 in our analysis.

26 Consider the following homogenization problem:

$$du_\varepsilon = \left[\operatorname{div} \left(A \left(\frac{x}{\varepsilon} \right) \nabla u_\varepsilon \right) - u_\varepsilon + f \left(\frac{x}{\varepsilon}, u_\varepsilon \right) \right] dt + \sigma \left(\frac{x}{\varepsilon}, u_\varepsilon \right) dW(t). \quad (1.16)$$

27 Here $x \in \mathbb{R}^d$, $u \in L^2_\rho(\mathbb{R}^d)$ with

$$\rho(x) = e^{-\kappa|x|},$$

28 $A(x)$ is a measurable periodic and uniformly elliptic matrix, $f(x, u)$ and $\sigma(x, u)$ are
29 uniformly bounded, continuous, periodic functions in x and Lipschitz in u . Assume
30 that the Lipschitz constant L and the parameter κ are small enough (more precisely,
31 they satisfy the smallness conditions (4.4), (4.6)–(4.8) below). Similar conditions
32 may be formulated for

$$\rho(x) = \frac{1}{1 + \kappa|x|^r}, \quad \kappa > 0, \quad r > d.$$

33 Along with (1.16), we consider the homogenized problem

$$du_0 = [\operatorname{div}(A_0 \nabla u_0) - u_0 + f_0(u_0)]dt + \sigma_0(u_0)dW(t), \quad (1.17)$$

1 where A_0 is the homogenized matrix described in [43, Sec. 1.2].

$$f_0(u) := \int_{\Pi} f(x, u) dx, \quad \sigma_0(u) := \int_{\Pi} \sigma(x, u) dx,$$

2 with $\Pi = [0, 1]^d$.

3 **Theorem 1.5.** *Under the above assumptions on A , ρ , f and σ , for every $t \geq 0$ we*
 4 *have*

$$\mathbb{E} \|u_\varepsilon(t) - u_0(t)\|_\rho^2 \rightarrow 0, \quad \varepsilon \rightarrow 0,$$

5 where u_ε and u_0 are the corresponding unique stationary solutions of (1.16)
 6 and (1.17), whose existence follows from Theorem 1.1.

7 Finally, we note that a similar convergence result holds in bounded domains
 8 with Dirichlet boundary conditions. In particular, for bounded $G \subset \mathbb{R}^d$, let u_ε be
 9 the unique stationary solution of

$$\begin{cases} du_\varepsilon = \left[\operatorname{div} \left(A \left(\frac{x}{\varepsilon} \right) \nabla u_\varepsilon \right) + f \left(\frac{x}{\varepsilon}, u_\varepsilon \right) \right] dt + \sigma \left(\frac{x}{\varepsilon}, u_\varepsilon \right) dW(t), & x \in G, \\ u = 0 & \text{on } \partial G. \end{cases} \quad (1.18)$$

10 where $A(x)$, $f(x, u)$ and $\sigma(x, u)$ are the same as in Theorem 1.5. Assume the Lip-
 11 schitz constant L for f and σ satisfy the smallness assumptions (4.6)–(4.8) with
 12 possibly different C_ν and γ . Finally, let u_0 be the corresponding stationary solution
 13 of the limiting equation

$$\begin{cases} du_0 = [\operatorname{div}(A_0 \nabla u_0) + f_0(u_0)] dt + \sigma_0(u_0) dW(t), & x \in G, \\ u = 0 & \text{on } \partial G. \end{cases} \quad (1.19)$$

14 **Theorem 1.6.** *For every $t \geq 0$, we have*

$$\mathbb{E} \|u_\varepsilon(t) - u_0(t)\|_{L^2(G)}^2 \rightarrow 0, \quad \varepsilon \rightarrow 0.$$

15 2. Vanishing Dissipation Limit

16 In this section, we prove Theorem 1.2. The proof follows from the two lemmas
 17 below.

18 **Lemma 2.1.** *Under the conditions of Theorem 1.2, Eq. (1.7) has a unique station-*
 19 *ary solution u_ε^* , defined for $t \in \mathbb{R}$, such that*

$$\sup_{t \in \mathbb{R}} \mathbb{E} \|u_\varepsilon^*(t)\|_{L^2(\mathbb{R}^d)} \leq C < \infty.$$

20 *with C independent of ε .*

21 **Remark 2.1.** As we mentioned earlier, the semigroup S_ε for the operator $D_\varepsilon u :=$
 22 $\Delta u - \varepsilon u$ satisfies the estimate (1.4). Thus, for fixed $\varepsilon > 0$, the existence and unique-
 23 ness of a bounded stationary solution u_ε^* , defined for $t \in \mathbb{R}$, follows from Theo-
 24 rem 1.1. However, the novelty of the present result is that this stationary solution
 25 is bounded uniformly in ε .

O. Misiats, O. Stanzhytskyi & N. K. Yip

Proof. The desired solution will be constructed as a limit of an iteration scheme. Set $u_0 \equiv 0$ and let u_{n+1} be the unique stationary solution of

$$du_{n+1} = (\Delta u_{n+1} - \varepsilon u_{n+1})dt + \sigma(x, u_n(t))dW(t).$$

Arguing as in [31, Theorem 5], such u_{n+1} is defined correctly, since at every step we have $\sup_{t \in \mathbb{R}} \mathbb{E} \|\sigma(\cdot, u_n)\|_{L^2(\mathbb{R}^d)}^2 < \infty$. Indeed, $\mathbb{E} \|\sigma(\cdot, 0)\|^2 < C$, hence there exists $u_1(t, x)$ satisfying

$$\sup_{t \in \mathbb{R}} \mathbb{E} \|u_1(t)\|_{L^2(\mathbb{R}^d)} < \infty.$$

Then, using the Lipschitz property of σ

$$\mathbb{E} \|\sigma(\cdot, u_1)\|_{L^2(\mathbb{R}^d)}^2 \leq 2 \int_{\mathbb{R}^d} |\sigma(x, 0)|^2 dx + 2L^2 \int_{\mathbb{R}^d} \mathbb{E} |u_1(t, x)|^2 dx < \infty$$

hence u_2 is defined correctly, and so on. Write

$$\begin{aligned} u_{n+1}(t) &= \int_{-\infty}^t S_\varepsilon(t-s) \sigma(\cdot, u_n) dW(s) \\ &= \sum_k \sqrt{a_k} \int_{-\infty}^t e^{-\varepsilon(t-s)} \int_{\mathbb{R}^d} G(t-s, x-y) \sigma(y, u_n(s, y)) e_k dy d\beta_k(s). \end{aligned}$$

Then

$$\begin{aligned} &\mathbb{E} \|u_{n+1}(t)\|_{L^2(\mathbb{R}^d)}^2 \\ &= \mathbb{E} \int_{\mathbb{R}^d} \left| \sum_k \sqrt{a_k} \int_{-\infty}^t e^{-\varepsilon(t-s)} \int_{\mathbb{R}^d} G(t-s, x-y) \sigma(y, u_n) e_k(y) dy d\beta_k(s) \right|^2 dx \\ &\leq \sum_k a_k \int_{\mathbb{R}^d} \int_{-\infty}^t e^{-2\varepsilon(t-s)} \mathbb{E} \left| \int_{\mathbb{R}^d} G(t-s, x-y) \sigma(y, u_n) e_k(y) dy \right|^2 ds dx. \end{aligned}$$

We proceed with the change variables in the k th term in the above summation:

$$\begin{aligned} &\int_{\mathbb{R}^d} \int_{-\infty}^t e^{-2\varepsilon(t-s)} \mathbb{E} \left| \int_{\mathbb{R}^d} G(t-s, x-y) \sigma(y, u_n) e_k(y) dy \right|^2 ds dx \\ &\leq \int_0^\infty ds \int_{\mathbb{R}^d} dx \mathbb{E} \left(\int_{\mathbb{R}^d} G(s, x-y) \sigma(y, u_n(t-s, y)) e_k(y) dy \right)^2 \\ &\leq \int_0^\infty ds \int_{\mathbb{R}^d} dx \mathbb{E} \left(\int_{\mathbb{R}^d} G(s, x-y) (\sigma(y, 0) + L|u_n(t-s, y)|) |e_k(y)| dy \right)^2 \\ &\leq 2 \int_0^\infty ds \int_{\mathbb{R}^d} dx \left(\int_{\mathbb{R}^d} G(s, x-y) |\sigma(y, 0)| |e_k(y)| dy \right)^2 \\ &\quad + 2 \int_0^\infty ds \int_{\mathbb{R}^d} dx \mathbb{E} \left(\int_{\mathbb{R}^d} G(s, x-y) L|u_n(t-s, y)| |e_k(y)| dy \right)^2 \\ &:= 2J_1 + 2J_2. \end{aligned}$$

1 To estimate the first integral, we write

$$\begin{aligned} J_1 &:= \int_0^\infty ds \int_{\mathbb{R}^d} dx \left(\int_{\mathbb{R}^d} G(s, x-y) |\sigma(y, 0)| |e_k(y)| dy \right)^2 \\ &= \int_0^1 [\cdots] ds + \int_1^\infty [\cdots] ds \\ &:= J_{11} + J_{12}. \end{aligned}$$

2 Since $|e_k(y)| \leq 1$, we immediately get

$$J_{11} \leq \int_0^1 \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} G(s, x-y) dx |\sigma(y, 0)|^2 dy \leq \int_{\mathbb{R}^d} |\sigma(y, 0)|^2 dy := C_\sigma.$$

3 As for the estimate for J_{12} , we have

$$\begin{aligned} J_{12} &\leq \int_1^\infty ds \int_{\mathbb{R}^d} dx \int_{\mathbb{R}^d} G^2(s, x-y) e_k^2(y) |\sigma(y, 0)|^2 dy \\ &\leq \int_{\mathbb{R}^d} |\sigma(y, 0)|^2 dy \cdot \int_1^\infty \int_{\mathbb{R}^d} G^2(s, x) dx ds \\ &:= C_\sigma C_d, \end{aligned}$$

4 where

$$C_d := \int_1^\infty \int_{\mathbb{R}^d} G^2(s, x) dx ds = \frac{1}{(8\pi)^{d/2} \left(\frac{d}{2} - 1\right)}, \quad d \geq 3.$$

5 Similarly, we write

$$J_2 = \int_0^\infty [\cdots] ds = \int_0^1 [\cdots] ds + \int_1^\infty [\cdots] ds := J_{21} + J_{22},$$

6 with

$$\begin{aligned} J_{21} &\leq \int_0^1 ds \int_{\mathbb{R}^d} dx \int_{\mathbb{R}^d} G(s, x-y) dy \int_{\mathbb{R}^d} G(s, x-y) L^2 \mathbb{E} |u_n(t-s, y)|^2 |e_k(y)|^2 dy \\ &\leq L^2 \sup_{t \in \mathbb{R}} \mathbb{E} \|u_n(t)\|_{L^2(\mathbb{R}^d)}^2 \end{aligned}$$

7 and

$$\begin{aligned} J_{22} &\leq L^2 \int_1^\infty \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} G^2(s, x-y) |e_k(y)|^2 dx dy \mathbb{E} \int_{\mathbb{R}^d} |u_n(t-s, y)|^2 dy \\ &\leq L^2 C_d \sup_{t \in \mathbb{R}} \mathbb{E} \|u_n(t)\|_{L^2(\mathbb{R}^d)}^2. \end{aligned}$$

8 Combining the estimates above, we have

$$\begin{aligned} &\int_{-\infty}^t \int_{\mathbb{R}^d} e^{-2\varepsilon(t-s)} \mathbb{E} \left| \int_{\mathbb{R}^d} G(t-s, x-y) \sigma(y, u_n) e_k(y) \right|^2 ds \\ &\leq 2(C_\sigma + C_\sigma C_d) + 2L^2(1 + C_d) \sup_{t \in \mathbb{R}} \mathbb{E} \|u_n(t)\|_{L^2(\mathbb{R}^d)}^2 \end{aligned}$$

O. Misiats, O. Stanzhytskyi & N. K. Yip

1 and hence, summing up in k , we have

$$\sup_{t \in \mathbb{R}} \mathbb{E} \|u_{n+1}(t)\|_{L^2(\mathbb{R}^d)}^2 \leq 2a(C_\sigma + C_\sigma C_d) + 2aL^2(1 + C_d) \sup_t \mathbb{E} \|u_n(t)\|_{L^2(\mathbb{R}^d)}^2.$$

2 Consequently,

$$\sup_{t \in \mathbb{R}} \mathbb{E} \|u_{n+1}(t)\|_{L^2(\mathbb{R}^d)}^2 \leq \frac{2aC_\sigma(1 + C_d)}{1 - 2a(1 + C_d)L^2} := C, \quad (2.1)$$

3 provided L satisfies (1.9).

4 We next proceed with establishing the convergence of u_n .

$$\begin{aligned} & \mathbb{E} \|u_{n+1}(t) - u_n(t)\|_{L^2(\mathbb{R}^d)}^2 \\ & \leq \sum_k a_k \int_{\mathbb{R}^d} \int_{-\infty}^t e^{-2\varepsilon(t-s)} \mathbb{E} \left(\int_{\mathbb{R}^d} G(t-s, x-y) \right. \\ & \quad \times [\sigma(y, u_n(y)) - \sigma(y, u_{n-1}(y))] e_k(y) dy \Big)^2 ds dx \\ & \leq L^2 \sum_k a_k \int_{\mathbb{R}^d} \int_0^\infty \mathbb{E} \left(\int_{\mathbb{R}^d} G(s, x-y) |u_n(t-s, y) - u_{n-1}(t-s, y)| \right. \\ & \quad \times |e_k(y)| dy \Big)^2 ds dx. \end{aligned}$$

5 Once again, fix $k \geq 1$. We have

$$\begin{aligned} & \int_0^\infty \int_{\mathbb{R}^d} \mathbb{E} \left(\int_{\mathbb{R}^d} G(s, x-y) |u_n(t-s, y) - u_{n-1}(t-s, y)| |e_k(y)| dy \right)^2 dx ds \\ & = \int_0^1 [\cdots] ds + \int_1^\infty [\cdots] ds := I_1 + I_2. \end{aligned}$$

6 Analogous estimates yield

$$I_1 \leq \sup_{t \in \mathbb{R}} \mathbb{E} \|u_n(t) - u_{n-1}(t)\|_{L^2(\mathbb{R}^d)}^2$$

7 and

$$I_2 \leq C_d \sup_{t \in \mathbb{R}} \mathbb{E} \|u_n(t) - u_{n-1}(t)\|_{L^2(\mathbb{R}^d)}^2.$$

8 Thus

$$\mathbb{E} \|u_{n+1}(t) - u_n(t)\|_{L^2(\mathbb{R}^d)}^2 \leq L^2 a(1 + C_d) \sup_t \mathbb{E} \|u_n(t) - u_{n-1}(t)\|_{L^2(\mathbb{R}^d)}^2. \quad (2.2)$$

9 If L satisfies (1.9), iterating (2.2) we deduce that u_n is a Cauchy sequence in $L^2(\mathbb{R}^d)$.

10 Thus, there exists a limiting function $u_\varepsilon^*(t, \cdot) \in L^2(\mathbb{R}^d)$. It is straightforward to
11 verify (see, e.g. [31, Theorem 3]) that $u_\varepsilon^*(t, \cdot)$ is a mild solution to (1.7) and satisfies

$$\sup_{t \in \mathbb{R}} \mathbb{E} \|u_\varepsilon^*(t)\|_{L^2(\mathbb{R}^d)}^2 \leq C < \infty.$$

12 where C is given by (2.1) and is independent of ε . □

1 **Lemma 2.2.** *There exists $u_0(t)$, which solves (1.8), such that for all $t \geq 0$*

$$\mathbb{E}\|u_\varepsilon^*(t) - u_0(t)\|_{L^2(\mathbb{R}^d)} \rightarrow 0, \quad \varepsilon \rightarrow 0,$$

2 *where u_ε^* is the solution constructed in Lemma 2.1.*

3 **Proof.** We first show that for fixed $t \geq 0$, $u_\varepsilon^*(t)$ is a Cauchy sequence in ε . For
4 $0 < \varepsilon < \varepsilon_1$, we have

$$\begin{aligned} & \mathbb{E}\|u_\varepsilon^*(t) - u_{\varepsilon_1}^*(t)\|_{L^2(\mathbb{R}^d)}^2 \\ &= \sum_k a_k \int_{\mathbb{R}^d} \mathbb{E} \int_{-\infty}^t \left[\int_{\mathbb{R}^d} [e^{-\varepsilon(t-s)} G\sigma(u_\varepsilon^*)e_k - e^{-\varepsilon_1(t-s)} G\sigma(u_{\varepsilon_1}^*)e_k] dy \right]^2 ds dx \end{aligned}$$

5 Adding and subtracting $e^{-\varepsilon_1(t-s)} G\sigma(u_\varepsilon^*)e_k$ to each term in the summation above,
6 we get

$$\begin{aligned} & \mathbb{E} \int_{\mathbb{R}^d} \int_{-\infty}^t \left[\int_{\mathbb{R}^d} [e^{-\varepsilon(t-s)} G\sigma(u_\varepsilon^*)e_k - e^{-\varepsilon_1(t-s)} G\sigma(u_{\varepsilon_1}^*)e_k] dy \right]^2 ds dx \\ & \leq 2\mathbb{E} \int_{\mathbb{R}^d} \int_{-\infty}^t \left[(e^{-\varepsilon(t-s)} - e^{-\varepsilon_1(t-s)}) \int_{\mathbb{R}^d} G\sigma(u_\varepsilon^*)e_k dy \right]^2 ds dx \\ & \quad + 2\mathbb{E} \int_{\mathbb{R}^d} \int_{-\infty}^t \left[e^{-\varepsilon_1(t-s)} \int_{\mathbb{R}^d} G[\sigma(u_\varepsilon^*) - \sigma(u_{\varepsilon_1}^*)]e_k dy \right]^2 ds dx \\ & := 2S_1 + 2S_2. \end{aligned} \tag{2.3}$$

7 After the change of variables in t , the first term in (2.3) can be further estimated
8 as follows:

$$\begin{aligned} S_1 & \leq \int_{\mathbb{R}^d} \int_0^\infty [e^{-\varepsilon s} - e^{-\varepsilon_1 s}]^2 \mathbb{E} \left[\int_{\mathbb{R}^d} G(s, x-y) |\sigma(y, 0) + Lu_\varepsilon^*(t-s, y)| |e_k| dy \right]^2 ds dx \\ & \leq 2 \int_{\mathbb{R}^d} \int_0^\infty [e^{-\varepsilon s} - e^{-\varepsilon_1 s}]^2 \mathbb{E} \left[\int_{\mathbb{R}^d} G(s, x-y) |\sigma(y, 0)| |e_k| dy \right]^2 ds dx \\ & \quad + 2L^2 \int_{\mathbb{R}^d} \int_0^\infty [e^{-\varepsilon s} - e^{-\varepsilon_1 s}]^2 \mathbb{E} \left[\int_{\mathbb{R}^d} G(s, x-y) |u_\varepsilon^*(t-s, y)| |e_k| dy \right]^2 ds dx \\ & := S_{11} + S_{12}. \end{aligned}$$

9 Using the bound $\sup_t \mathbb{E}\|u_\varepsilon^*\|^2 \leq C < \infty$, by Dominated Convergence Theorem,
10 $S_{11} \rightarrow 0$ and $S_{12} \rightarrow 0$ as $\varepsilon \rightarrow 0$, $\varepsilon_1 \rightarrow 0$. It remains to estimate S_2 . A calculation,
11 analogous to (2.2), yields

$$S_2 \leq (1 + C_d) L^2 \sup_{t \in \mathbb{R}} \mathbb{E}\|u_\varepsilon^*(t) - u_{\varepsilon_1}^*(t)\|_{L^2(\mathbb{R}^d)}^2.$$

12 Altogether,

$$\mathbb{E}\|u_\varepsilon^*(t) - u_{\varepsilon_1}^*(t)\|_{L^2(\mathbb{R}^d)}^2 \leq \varphi(\varepsilon, \varepsilon_1) + 2aL^2(1 + C_d) \sup_{t \in \mathbb{R}} \mathbb{E}\|u_\varepsilon^*(t) - u_{\varepsilon_1}^*(t)\|_{L^2(\mathbb{R}^d)}^2,$$

O. Misiats, O. Stanzhytskyi & N. K. Yip

1 where $\varphi(\varepsilon, \varepsilon_1) \rightarrow 0, \varepsilon, \varepsilon_1 \rightarrow 0$. In other words,

$$\sup_{t \in \mathbb{R}} \mathbb{E} \|u_\varepsilon^*(t) - u_{\varepsilon_1}^*(t)\|_{L^2(\mathbb{R}^d)}^2 \leq \frac{\varphi(\varepsilon, \varepsilon_1)}{1 - 2aL^2(1 + C_d)} \rightarrow 0, \quad \varepsilon, \varepsilon_1 \rightarrow 0.$$

2 Thus there is u_0^* such that $u_\varepsilon^* \rightarrow u_0^*$ as $\varepsilon \rightarrow 0$.

3 It remains to show that the sequence u_ε^* converges to the solution of the limiting
4 equation

$$\begin{cases} du_0 = \Delta u_0 dt + \sigma(x, u_0) dW(t), \\ u_0(0) = u_0. \end{cases}$$

5 Denoting

$$S_0(t)u(\cdot) := \int_{\mathbb{R}^d} G(t, x - y)u(y)dy,$$

6 we have

$$\begin{aligned} & \mathbb{E} \|u_\varepsilon^*(t) - u^0(t)\|_{L^2(\mathbb{R}^d)}^2 \\ & \leq 3\mathbb{E} \|u_\varepsilon^*(0) - u^0\|_{L^2(\mathbb{R}^d)}^2 + \varepsilon^2 \mathbb{E} \left\| \int_0^t S_0(t-s)u_\varepsilon(s)ds \right\|_{L^2(\mathbb{R}^d)}^2 \\ & \quad + \mathbb{E} \left\| \int_0^t S_0(t-s)[\sigma(\cdot, u_\varepsilon^*) - \sigma(\cdot, u^0)]dW(s) \right\|_{L^2(\mathbb{R}^d)}^2 \\ & \leq 3E \|u_\varepsilon^*(0) - u^0\|_{L^2(\mathbb{R}^d)}^2 + B\varepsilon^2 + L^2a \int_0^t \mathbb{E} \|u_\varepsilon^*(s) - u^0(s)\|_{L^2(\mathbb{R}^d)}^2 ds \end{aligned}$$

7 for some constant $B > 0$. By Gronwall's inequality

$$\mathbb{E} \|u_\varepsilon^*(t) - u^0(t)\|_{L^2(\mathbb{R}^d)}^2 \rightarrow 0, \quad \varepsilon \rightarrow 0.$$

8 In particular, since the stationary solutions u_ε^* strongly converge to u^0 , u^0 is also
9 stationary. This concludes the proof of the lemma, and thus the statement of the
10 Theorem 1.2 follows. $\square$

11 3. Perturbation Analysis. Proof of Theorem 1.4

12 Let $\lambda_1^\varepsilon < 0$ and $\lambda_1 < 0$ be the principle eigenvalues of $D_\varepsilon u := \Delta u - qu - \varepsilon q_1 u$
13 and $Du := \Delta u - qu$, respectively. Choose $\lambda := \frac{\lambda_1}{2} < 0$. Since $\lambda_1^\varepsilon = \lambda_1 + o_1(\varepsilon)$, we
14 have $\lambda_1^\varepsilon < \lambda$ for sufficiently small ε . Denote $S_\varepsilon(t)$ and $S(t)$ to be the corresponding
15 semigroups for D_ε and D . Then

$$\|S_\varepsilon(t)\|_{\mathcal{L}} \leq e^{\lambda t}, \quad \|S(t)\|_{\mathcal{L}} \leq e^{\lambda t}, \quad (3.1)$$

16 where $\|D\|_{\mathcal{L}}$ stands for the operator norm of D . By Theorem 1.1, (1.12) and (1.13)
have unique stationary solutions, denoted with u_ε and u respectively, defined for

1 $t \in \mathbb{R}$. Moreover, arguing as in the proof of Lemma 2.1, it is straightforward to
 2 verify that

$$\sup_{t \in \mathbb{R}} \mathbb{E} \|u_\varepsilon(t)\|_{L^2(\mathbb{R}^d)}^2 \leq \frac{\frac{4}{\lambda^2} \|f(\cdot, 0)\|_{L^2(\mathbb{R}^d)}^2 - \frac{4a}{\lambda} \|\sigma(\cdot, 0)\|_{L^2(\mathbb{R}^d)}^2}{1 - L^2 \left(\frac{4}{\lambda^2} - \frac{2a}{\lambda} \right)} := C$$

3 provided L satisfies the smallness condition (1.14). The following lemma establishes
 4 the convergence properties of the semigroup.

5 **Lemma 3.1.** *Assume q and q_1 satisfy the assumptions of Theorem 1.3. Then*

6 (i) *For any $u_0 \in L^2(\mathbb{R}^d)$ and $t \geq 0$ we have*

$$\|S_\varepsilon(t)u_0 - S(t)u_0\|_{L^2(\mathbb{R}^d)} \rightarrow 0, \quad \varepsilon \rightarrow 0.$$

7 (ii) *If, in addition, $q_1 \in L^\infty(\mathbb{R}^d)$, then*

$$\|S_\varepsilon(t)u_0 - S(t)u_0\|_{L^2(\mathbb{R}^d)} \leq c\lambda\varepsilon \|q_1\|_{L^\infty(\mathbb{R}^d)} e^{\frac{\lambda}{2}t} \|u_0\|_{L^2(\mathbb{R}^d)}.$$

8 **Proof.** We start with the proof of (i). Write

$$u_0 = \sum_{k=1}^{\infty} c_k \Psi_k = \sum_{k=1}^{\infty} c_k^\varepsilon \Psi_k^\varepsilon$$

9 with

$$c_k = \int_{\mathbb{R}^d} u_0(x) \Psi_k(x) dx \quad \text{and} \quad c_k^\varepsilon = \int_{\mathbb{R}^d} u_0(x) \Psi_k^\varepsilon(x) dx.$$

10 In particular,

$$|c_k - c_k^\varepsilon| \leq \int_{\mathbb{R}^d} |\Psi_k^\varepsilon(x) - \Psi_k(x)| |u_0(x)| dx \leq O(\varepsilon) \|u_0\|_{L^2(\mathbb{R}^d)}.$$

11 Then

$$\begin{aligned} & \|S_\varepsilon(t)u_0 - S(t)u_0\|_{L^2(\mathbb{R}^d)}^2 \\ &= \left\| \sum_{k=1}^{\infty} e^{\lambda_k^\varepsilon t} c_k^\varepsilon \Psi_k^\varepsilon - e^{\lambda_k t} c_k \Psi_k \right\|_{L^2(\mathbb{R}^d)}^2 \\ &\leq 2 \left\| \sum_{k=1}^N e^{\lambda_k^\varepsilon t} c_k^\varepsilon \Psi_k^\varepsilon - e^{\lambda_k t} c_k \Psi_k \right\|_{L^2(\mathbb{R}^d)}^2 + 2 \left\| \sum_{k=N}^{\infty} e^{\lambda_k^\varepsilon t} c_k^\varepsilon \Psi_k^\varepsilon - e^{\lambda_k t} c_k \Psi_k \right\|_{L^2(\mathbb{R}^d)}^2. \end{aligned} \tag{3.2}$$

12 Fix $\mu > 0$. For this μ we can find $N > 0$ such that

$$\left\| \sum_{k=N+1}^{\infty} e^{\lambda_k t} c_k \Psi_k \right\|_{L^2(\mathbb{R}^d)}^2 \leq \sum_{k=N+1}^{\infty} |c_k|^2 \leq \frac{\mu}{4},$$

O. Misiats, O. Stanzhytskyi & N. K. Yip

1 as well as

$$e^{2\lambda_N t} \|u_0\|_{L^2(\mathbb{R}^d)}^2 \leq \frac{\mu}{8}. \quad (3.3)$$

2 Since

$$\left\| \sum_{k=N+1}^{\infty} e^{\lambda_k^\varepsilon t} c_k^\varepsilon \Psi_k^\varepsilon \right\|_{L^2(\mathbb{R}^d)}^2 \leq e^{2\lambda_N^\varepsilon t} \|u_0\|_{L^2(\mathbb{R}^d)}^2,$$

3 from (3.3) and the convergence $\lambda_N^\varepsilon \rightarrow \lambda_N$ we may conclude that there is $\varepsilon_0 > 0$
4 such that for $\varepsilon < \varepsilon_0$

$$\left\| \sum_{k=N+1}^{\infty} e^{\lambda_k^\varepsilon t} c_k^\varepsilon \Psi_k^\varepsilon \right\|_{L^2(\mathbb{R}^d)}^2 \leq \frac{\mu}{4},$$

5 thus the second term in (3.2) is less than $\frac{\mu}{2}$. As for the first term in (3.2), we have

$$\begin{aligned} & \left\| \sum_{k=1}^N e^{\lambda_k^\varepsilon t} c_k^\varepsilon \Psi_k^\varepsilon - e^{\lambda_k t} c_k \Psi_k \right\|_{L^2(\mathbb{R}^d)}^2 \\ & \leq 2 \left\| \sum_{k=1}^N e^{\lambda_k t} c_k (\Psi_k^\varepsilon - \Psi_k) \right\|_{L^2(\mathbb{R}^d)}^2 + 2 \left\| \sum_{k=1}^N (e^{\lambda_k^\varepsilon t} c_k^\varepsilon - e^{\lambda_k t} c_k) \Psi_k^\varepsilon \right\|_{L^2(\mathbb{R}^d)}^2 \\ & \leq 2N \sum_{k=1}^N e^{2\lambda_k t} |c_k|^2 \|\Psi_k^\varepsilon - \Psi_k\|_{L^2(\mathbb{R}^d)}^2 + 2N \sum_{k=1}^N |e^{\lambda_k^\varepsilon t} c_k^\varepsilon - e^{\lambda_k t} c_k|^2 \leq \frac{\mu}{2} \end{aligned}$$

6 for ε small enough. Thus, we have (i). To show (ii), note that

$$\begin{aligned} & \|S_\varepsilon(t)u_0 - S_0(t)u_0\|_{L^2(\mathbb{R}^d)} \\ & = \left\| \int_0^t S_0(t-s) \varepsilon q_1(\cdot) u_\varepsilon(s) ds \right\|_{L^2(\mathbb{R}^d)} \\ & \leq \varepsilon \|q_1\|_{L^\infty(\mathbb{R}^d)} \|u_0\|_{L^2(\mathbb{R}^d)} \int_0^t e^{\lambda(t-s)+\lambda s} ds = \varepsilon \|q_1\|_{L^\infty(\mathbb{R}^d)} \|u_0\|_{L^2(\mathbb{R}^d)} t e^{\lambda t} \\ & \leq \varepsilon C \lambda \|q_1\|_{L^\infty(\mathbb{R}^d)} \|u_0\|_{L^2(\mathbb{R}^d)} e^{\frac{\lambda}{2} t}. \quad \square \end{aligned}$$

7 Introduce

$$u_\varepsilon^n := \int_{-n}^t S_\varepsilon(t-s) f(\cdot, u_\varepsilon(s)) ds + \int_{-n}^t S_\varepsilon(t-s) \sigma(\cdot, u_\varepsilon(s)) dW(s)$$

8 and

$$u^n := \int_{-n}^t S(t-s) f(\cdot, u(s)) ds + \int_{-n}^t S(t-s) \sigma(\cdot, u(s)) dW(s)$$

1 Then, for fixed $t \in \mathbb{R}$

$$\begin{aligned}
& \mathbb{E} \|u_\varepsilon(t) - u_\varepsilon^n(t)\|_{L^2(\mathbb{R}^d)}^2 \\
& \leq 2 \left[E \left\| \int_{-\infty}^{-n} S_\varepsilon(t-s) f(\cdot, u_\varepsilon(s)) ds \right\|_{L^2(\mathbb{R}^d)}^2 \right. \\
& \quad \left. + E \left\| \int_{-\infty}^{-n} S_\varepsilon(t-s) \sigma(\cdot, u_\varepsilon(s)) dW(s) \right\|_{L^2(\mathbb{R}^d)}^2 \right] \\
& \leq 4 \left(\int_{-\infty}^{-n} e^{\lambda(t-s)} ds \right) \int_{-\infty}^{-n} e^{\lambda(t-s)} (\|f(\cdot, 0)\|_{L^2(\mathbb{R}^d)}^2 + L^2 \mathbb{E} \|u_\varepsilon\|_{L^2(\mathbb{R}^d)}^2) ds \\
& \quad + 4a \int_{-\infty}^{-n} e^{2\lambda(t-s)} (\|\sigma(\cdot, 0)\|_{L^2(\mathbb{R}^d)}^2 + L^2 \mathbb{E} \|u_\varepsilon\|_{L^2(\mathbb{R}^d)}^2) ds \\
& \leq 4(\|f(\cdot, 0)\|_{L^2(\mathbb{R}^d)}^2 + L^2 C) \left(\int_{-\infty}^{-n} e^{\lambda(t-s)} ds \right)^2 \\
& \quad + 4a(\|\sigma(\cdot, 0)\|_{L^2(\mathbb{R}^d)}^2 + L^2 C) \int_{-\infty}^{-n} e^{\lambda(t-s)} ds \\
& \leq B_1 e^{2\lambda(t+n)} \rightarrow 0, \quad n \rightarrow \infty,
\end{aligned} \tag{3.4}$$

2 with B_1 independent of ε . Similarly,

$$\mathbb{E} \|u(t) - u^n(t)\|_{L^2(\mathbb{R}^d)}^2 \leq B_1 e^{2\lambda(t+n)} \rightarrow 0, \quad n \rightarrow \infty.$$

3 For fixed $t \geq 0$, we have

$$\begin{aligned}
& \mathbb{E} \|u_\varepsilon^n(t) - u^n(t)\|_{L^2(\mathbb{R}^d)}^2 \\
& \leq 2 \mathbb{E} \left\| \int_{-n}^t (S_\varepsilon(t-s) f(\cdot, u_\varepsilon(s)) - S(t-s) f(\cdot, u(s))) ds \right\|_{L^2(\mathbb{R}^d)}^2 \\
& \quad + 2 \mathbb{E} \left\| \int_{-n}^t (S_\varepsilon(t-s) \sigma(\cdot, u_\varepsilon(s)) - S(t-s) \sigma(\cdot, u(s))) dW(s) \right\|_{L^2(\mathbb{R}^d)}^2 \\
& := 2(J_1 + J_2).
\end{aligned} \tag{3.5}$$

4 Adding and subtracting $S_\varepsilon(t-s) f(\cdot, u_\varepsilon^n(s))$ and $S(t-s) f(\cdot, u^n(s))$, and using the
 5 contraction property (3.1) together with the Lipschitz property of f , we have

$$\begin{aligned}
J_1 & \leq \mathbb{E} \left[\int_{-n}^t (e^{\lambda(t-s)} L \|u_\varepsilon(s) - u_\varepsilon^n(s)\|_{L^2(\mathbb{R}^d)} + e^{\lambda(t-s)} L \|u(s) - u^n(s)\|_{L^2(\mathbb{R}^d)} \right. \\
& \quad \left. + \|S_\varepsilon(t-s) f(\cdot, u_\varepsilon^n(s)) - S(t-s) f(\cdot, u^n(s))\|_{L^2(\mathbb{R}^d)} ds \right]^2
\end{aligned}$$

O. Misiats, O. Stanzhytskyi & N. K. Yip

$$\begin{aligned}
&\leq 3\mathbb{E} \left(\int_{-n}^t e^{\lambda(t-s)} L \|u_\varepsilon(s) - u_\varepsilon^n(s)\|_{L^2(\mathbb{R}^d)} ds \right)^2 \\
&\quad + 3\mathbb{E} \left(\int_{-n}^t e^{\lambda(t-s)} L \|u(s) - u^n(s)\|_{L^2(\mathbb{R}^d)} ds \right)^2 \\
&\quad + 3\mathbb{E} \left(\int_{-n}^t \|S_\varepsilon(t-s)f(\cdot, u_\varepsilon^n(s)) - S(t-s)f(\cdot, u^n(s))\|_{L^2(\mathbb{R}^d)} ds \right)^2 \\
&:= 3(J_{11} + J_{12} + J_{13}).
\end{aligned}$$

1 By (3.4) and Cauchy–Schwartz, we have

$$\begin{aligned}
J_{11} &\leq \int_{-n}^t e^{\lambda(t-s)} ds \cdot \int_{-n}^t L^2 e^{\lambda(t-s)} \|u(s) - u^n(s)\|_{L^2(\mathbb{R}^d)}^2 ds \\
&\leq -\frac{1}{\lambda} \int_{-n}^t e^{\lambda(t-s)} L^2 B_1 e^{2\lambda(s+n)} ds = -B_1 \frac{L^2}{\lambda} e^{\lambda(2n+t)} \int_{-n}^t e^{\lambda s} ds \\
&\leq B_1 \frac{L^2}{\lambda^2} e^{\lambda(t+n)}.
\end{aligned}$$

2 Similarly,

$$J_{12} \leq B_1 \frac{L^2}{\lambda^2} e^{\lambda(t+n)}.$$

3 Finally,

$$\begin{aligned}
J_{13} &\leq \mathbb{E} \left[\int_{-n}^t \|S_\varepsilon(t-s)(f(\cdot, u_\varepsilon^n) - f(\cdot, u^n))\|_{L^2(\mathbb{R}^d)} ds \right. \\
&\quad \left. + \int_{-n}^t \|S_\varepsilon(t-s)(f(\cdot, u^n) - S(t-s)f(\cdot, u^n))\|_{L^2(\mathbb{R}^d)} ds \right]^2 \\
&\leq 2(J_{131} + J_{132})
\end{aligned}$$

4 We have

$$\begin{aligned}
J_{131} &\leq \mathbb{E} \left(\int_{-n}^t e^{\lambda(t-s)} L \|u_\varepsilon^n - u^n\|_{L^2(\mathbb{R}^d)} ds \right)^2 \\
&\leq -\frac{L^2}{\lambda} \int_{-n}^t e^{\lambda(t-s)} \mathbb{E} \|u_\varepsilon^n(s) - u^n(s)\|_{L^2(\mathbb{R}^d)}^2 ds.
\end{aligned}$$

5 Also, by Dominated Convergence Theorem and Lemma 3.1

$$J_{132} := \varphi_1(t, n, \varepsilon) \rightarrow 0, \quad \varepsilon \rightarrow 0.$$

6 Altogether,

$$J_1 \leq \frac{6}{\lambda^2} e^{\lambda(t+n)} - \frac{6L^2}{\lambda} \int_{-n}^t e^{\lambda(t-s)} \mathbb{E} \|u_\varepsilon^n(s) - u^n(s)\|_{L^2(\mathbb{R}^d)}^2 ds + 6\varphi_1(t, n, \varepsilon). \quad (3.6)$$

1 We now proceed with the estimate for J_2 :

$$\begin{aligned}
 J_2 &\leq 3\mathbb{E} \left\| \int_{-n}^t S_\varepsilon(t-s)(\sigma(\cdot, u_\varepsilon(s)) - \sigma(\cdot, u_\varepsilon^n(s)))dW(s) \right\|_{L^2(\mathbb{R}^d)}^2 \\
 &\quad + 3\mathbb{E} \left\| \int_{-n}^t S_\varepsilon(t-s)(\sigma(\cdot, u_\varepsilon^n(s)) - S(t-s)(\sigma(\cdot, u^n(s))))dW(s) \right\|_{L^2(\mathbb{R}^d)}^2 \\
 &\quad + 3\mathbb{E} \left\| \int_{-n}^t S(t-s)(\sigma(\cdot, u^n(s)) - \sigma(\cdot, u(s)))dW(s) \right\|_{L^2(\mathbb{R}^d)}^2 \\
 &:= 3(J_{21} + J_{22} + J_{23}).
 \end{aligned}$$

2 Using (3.4), we have

$$\begin{aligned}
 J_{21} &\leq L^2 \sum_k a_k \int_{-n}^t e^{2\lambda(t-s)} \mathbb{E} \|u_\varepsilon(s) - u_\varepsilon^n(s)\|_{L^2(\mathbb{R}^d)}^2 ds \\
 &\leq L^2 a \int_{-n}^t e^{2\lambda(t-s)} B_1 e^{2\lambda(s+n)} ds = aL^2 B_1 e^{2\lambda(t+n)}(t+n) \rightarrow 0, \quad n \rightarrow \infty.
 \end{aligned}$$

3 A similar estimate holds for J_{23} . It remains to estimate J_{22} :

$$\begin{aligned}
 J_{22} &= \mathbb{E} \left\| \int_{-n}^t [S_\varepsilon(t-s)(\sigma(\cdot, u_\varepsilon^n) - \sigma(\cdot, u^n)) \right. \\
 &\quad \left. + (S_\varepsilon(t-s) - S(t-s))\sigma(\cdot, u^n)]dW(s) \right\|_{L^2(\mathbb{R}^d)}^2 \\
 &\leq 2\mathbb{E} \left\| \int_{-n}^t S_\varepsilon(t-s)[\sigma(\cdot, u_\varepsilon^n) - \sigma(\cdot, u^n)]dW(s) \right\|_{L^2(\mathbb{R}^d)}^2 \\
 &\quad + 2\mathbb{E} \left\| \int_{-n}^t [S_\varepsilon(t-s)\sigma(\cdot, u^n(s)) - S(t-s)\sigma(\cdot, u^n(s))]dW(s) \right\|_{L^2(\mathbb{R}^d)}^2 \\
 &:= 2(J_{221} + J_{222}).
 \end{aligned}$$

4 The first term is estimated by Cauchy–Schwart

$$J_{221} \leq -\frac{aL^2}{\lambda} \int_{-n}^t e^{\lambda(t-s)} \mathbb{E} \|u_\varepsilon^n(s) - u^n(s)\|_{L^2(\mathbb{R}^d)}^2 ds.$$

5 On the other hand,

$$\begin{aligned}
 &\int_{-n}^t [S_\varepsilon(t-s)\sigma(\cdot, u^n(s)) - S(t-s)\sigma(\cdot, u^n(s))]dW(s) \\
 &= \sum_k \sqrt{a_k} \int_{-n}^t [S_\varepsilon(t-s)\sigma(\cdot, u^n(s))e_k - S(t-s)\sigma(\cdot, u^n(s))e_k]d\beta_k(s),
 \end{aligned}$$

O. Misiats, O. Stanzhytskyi & N. K. Yip

1 thus

$$\begin{aligned} J_{222} &= \sum_k a_k \int_{-n}^t \mathbb{E} \|S_\varepsilon(t-s)\sigma(\cdot, u^n(s))e_k - S(t-s)\sigma(\cdot, u^n(s))e_k\|_{L^2(\mathbb{R}^d)}^2 ds \\ &:= \varphi_2(t, n, \varepsilon) \rightarrow 0 \end{aligned}$$

2 as $\varepsilon \rightarrow 0$ by the Dominated Convergence Theorem and Lemma 3.1. Altogether

$$J_2 \leq 3[B_2 e^{\lambda(t+n)} - \frac{2aL^2}{\lambda} \int_{-n}^t e^{\lambda(t-s)} \mathbb{E} \|u_\varepsilon^n(s) - u^n(s)\|_{L^2(\mathbb{R}^d)}^2 ds + 2\varphi_2(t, n, \varepsilon)]. \quad (3.7)$$

3 Combining (3.5)–(3.7), we have

$$\begin{aligned} &\mathbb{E} \|u_\varepsilon^n(t) - u^n(t)\|_{L^2(\mathbb{R}^d)}^2 \\ &\leq B_3 e^{\lambda(t+n)} + \varphi_3(t, n, \varepsilon) - \frac{12L^2(1+a)}{\lambda} \int_{-n}^t e^{\lambda(t-s)} \mathbb{E} \|u_\varepsilon^n(s) - u^n(s)\|_{L^2(\mathbb{R}^d)}^2 ds \end{aligned}$$

4 By Gronwall inequality, denoting $\gamma := -\frac{12L^2(1+a)}{\lambda} > 0$, we have

$$\begin{aligned} \mathbb{E} \|u_\varepsilon^n(t) - u^n(t)\|_{L^2(\mathbb{R}^d)}^2 &\leq B_3 e^{\lambda(t+n)} + \varphi_3(t, n, \varepsilon) \\ &\quad + \gamma \int_{-n}^t e^{\gamma(t-s)} e^{\lambda(t-s)} (B_3 e^{\lambda(s+n)} + \varphi_3(s, n, \varepsilon)) ds. \end{aligned}$$

5 The condition (1.15) implies that $\lambda + \gamma < 0$, thus we have

$$\int_{-n}^t e^{\gamma(t-s)} e^{\lambda(t-s)} e^{\lambda(s+n)} ds = \frac{1}{\gamma} [e^{(\lambda+\gamma)(t+n)} - e^{\lambda(t+n)}] \rightarrow 0, \quad n \rightarrow \infty.$$

6 To complete the proof of Theorem 1.4, fix $\delta > 0$ and choose $n \geq 1$ such that

$$2B_1 e^{2\lambda(t+n)} + B_3 e^{\lambda(t+n)} + \frac{B_3}{\gamma} [e^{(\lambda+\gamma)(t+n)} - e^{\lambda(t+n)}] < \frac{\delta}{2}.$$

7 For such n we may now choose $\varepsilon > 0$ such that

$$\varphi_3(t, n, \varepsilon) + \gamma \int_{-n}^t e^{\gamma(t-s)} e^{\lambda(t-s)} \varphi_3(s, n, \varepsilon) ds < \frac{\delta}{2}$$

8 and therefore

$$\mathbb{E} \|u_\varepsilon(t) - u(t)\|_{L^2(\mathbb{R}^d)}^2 \leq \frac{\delta}{2} + \frac{\delta}{2} = \delta,$$

9 which completes the proof.

10 4. Homogenization

11 4.1. Notation and preliminaries

12 Assume that $A(x)$ is uniformly elliptic

$$\nu |\xi|^2 \leq A(x) \xi \cdot \xi \leq \frac{1}{\nu} |\xi|^2, \quad 0 < \nu \leq 1, \quad \xi \in \mathbb{R}^d. \quad (4.1)$$

1 Denote $A_0(x)$ to be the homogenized matrix (see [43, Sec. 1.2] for the precise
 2 definition). It follows from linear homogenization theory that A_0 satisfies (4.1)
 3 given that $A(x)$ satisfies it. Next, let

- 4 • $K(x, y, t)$ be the fundamental solution of

$$u_t = \operatorname{div}(A(x)\nabla u);$$

- 5 • $K_\varepsilon(x, y, t)$ be the fundamental solution of

$$u_t = \operatorname{div}(A(x/\varepsilon)\nabla u);$$

- 6 • $K_0(x, y, t)$ be the fundamental solution of

$$u_t = \operatorname{div}(A_0(x)\nabla u).$$

•

$$S_\varepsilon(t)u := \int_{\mathbb{R}^d} K_\varepsilon(x, y, t)u(y)dy; \quad S_0(t)u := \int_{\mathbb{R}^d} K_0(x, y, t)u(y)dy.$$

7 A simple rescaling yields

$$K_\varepsilon(x, y, t) = \varepsilon^{-d} K\left(\frac{x}{\varepsilon}, \frac{y}{\varepsilon}, \frac{t}{\varepsilon^2}\right).$$

8 We also assume that $B_{f,\sigma} > 0$ is such that

$$|f(x, u)| \leq B_{f,\sigma} \quad \text{and} \quad |\sigma(x, u)| \leq B_{f,\sigma}.$$

9 Let $\Omega = B_R(0)$, $Y = \Pi = [0, 1]^d$. Following [1], denote $L^1[\Omega, C_\#(Y)]$ to be the space
 10 of functions, measurable and integrable in $x \in \Omega$, with values in the Banach space
 11 of continuous functions, Y -periodic in y . We are going to make use of the following
 12 lemmas to characterize the space $L^1[\Omega, C_\#(Y)]$.

13 **Lemma 4.1.** ([1, Lemma 5.3]) *A function $\psi(x, y)$ belongs to $L^1[\Omega, C_\#(Y)]$ if and*
 14 *only if there exists a subset E (independent of y) of measure zero in Ω such that*

- 15 (i) *For any $x \in \Omega \setminus E$, the function $y \rightarrow \psi(x, y)$ is continuous and Y -periodic;*
 16 (ii) *For any $y \in Y$, the function $x \rightarrow \psi(x, y)$ is measurable on Ω .*
 17 (iii) *The function $x \rightarrow \sup_y |\psi(x, y)|$ has a finite $L^1(\Omega)$ norm.*

18 **Lemma 4.2.** ([1, Lemma 5.2]) *If $\psi(x, y) \in L^1[\Omega, C_\#(Y)]$, then*

$$\int_{\Omega} \psi\left(x, \frac{x}{\varepsilon}\right) dx \rightarrow \int_{\Pi} \int_{\Omega} \psi(x, y) dx dy, \quad \varepsilon \rightarrow 0.$$

19 4.2. Nash–Aranson estimates. Semigroup estimates

20 **Lemma 4.3.** ([43, Chap. 2]) *There exists $C_N = C(d, \nu)$ such that*

$$0 \leq K(x, y, t) \leq C_N t^{-d/2} e^{-\frac{\nu|x-y|^2}{8t}};$$

$$0 \leq K_0(x, y, t) \leq C_N t^{-d/2} e^{-\frac{\nu|x-y|^2}{8t}}.$$

O. Misiats, O. Stanzhytskyi & N. K. Yip

1 If $S^{(m)}(t)u$ is the semigroup, generated by $\Delta u - mu$ in $L^2(\mathbb{R}^d)$, then $S^{(m)}(t)u =$
2 $e^{-mt}S^{(0)}(t)u$ and thus

$$\|S^{(m)}(t)u\|_{L^2(\mathbb{R}^d)} \leq e^{-mt}\|u\|_{L^2(\mathbb{R}^d)}. \quad (4.2)$$

3 An analog of this estimate holds in weighted spaces.

Lemma 4.4. ([15, Proposition 9.4.5]) *Let $u \in L^2_\rho(\mathbb{R}^d)$ with $\rho(x) = e^{-\kappa|x|}$. Then*

$$\|S^{(m)}(t)u\|_\rho \leq e^{(\kappa^2/2-m)t}\|u\|_\rho. \quad (4.3)$$

A similar estimate holds for $\rho(x) = \frac{1}{1+\kappa|x|^r}$, $\kappa > 0, r > d$.

6 Rescaling time $s = tm$, one can see that the estimates (4.2) and (4.3) are valid
7 for the semigroup, generated by $\frac{1}{m}\Delta u - u$ in $L^2(\mathbb{R}^d)$ and $L^2_\rho(\mathbb{R}^d)$, respectively.
8 Combining Lemmas 4.3 and 4.4, for any $\varphi \in L^2_\rho(\mathbb{R}^d)$ we have

$$\begin{aligned} \|S_\varepsilon(t)\varphi\|_\rho^2 &= \int_{\mathbb{R}^d} e^{-2t} \left| \int_{\mathbb{R}^d} K_\varepsilon(x, y, t) \varphi(y) dy \right|^2 \rho(x) dx \\ &\leq \int_{\mathbb{R}^d} e^{-2t} \left(\int_{\mathbb{R}^d} C_N t^{-d/2} e^{-\frac{|x-y|^2}{8t/\nu}} |\varphi(y)| dy \right)^2 \rho(x) dx \\ &= \left(\frac{8\pi}{\nu} \right)^d C_N^2 \int_{\mathbb{R}^d} e^{-2t} \left(\int_{\mathbb{R}^d} \frac{1}{(8\pi t/\nu)^{d/2}} e^{-\frac{|x-y|^2}{8t/\nu}} |\varphi| dy \right)^2 \rho(x) dx \\ &= \left(\frac{8\pi}{\nu} \right)^d C_N^2 \|S^{(\nu)}(t/\nu)\varphi(\cdot)\|_\rho^2 \leq \left(\frac{8\pi}{\nu} \right)^d C_N^2 e^{2(\kappa^2/2-\nu)t/\nu} \|\varphi\|_\rho^2. \end{aligned}$$

9 In other words, if

$$\kappa^2/2 < \nu, \quad (4.4)$$

10 we showed that

$$\|S_\varepsilon(t)\varphi\|_\rho \leq C_\nu e^{-\gamma t} \|\varphi\|_\rho, \quad (4.5)$$

11 with $\gamma := 1 - \frac{\kappa^2}{2\nu} > 0$ and $C_\nu^2 = \left(\frac{8\pi}{\nu}\right)^d C_N^2$. Similarly,

$$\|S_0(t)\varphi\|_\rho \leq C_\nu e^{-\gamma t} \|\varphi\|_\rho.$$

12 We may now apply Theorem 1.1 to conclude that, the Lipschits constant L satisfies
13 the smallness conditions [31]:

$$C_\nu^2 L^2 \frac{4 + 2\gamma a}{\gamma^2} < 1 \quad (4.6)$$

14 and

$$C_\nu^2 L^2 \frac{1 + \gamma a}{\gamma} < \frac{2}{3}, \quad (4.7)$$

15 then there exist unique stationary solutions u_ε and u_0 of (1.16) and (1.17), respec-
16 tively, defined for $t \in \mathbb{R}$, such that

$$\sup_{t \in \mathbb{R}} \mathbb{E} \|u_\varepsilon(t)\|_\rho^2 < C \quad \text{and} \quad \sup_{t \in \mathbb{R}} \mathbb{E} \|u_0(t)\|_\rho^2 < C.$$

AQ: Is this Proposition or Property? Kindly check.

1 Finally, assume that L satisfies one more smallness condition

$$12 \left(\frac{1}{\gamma} + a \right) C_\nu^2 L^2 < \gamma. \quad (4.8)$$

2 **4.3. Proof of Theorem 1.5**

3 For fixed $t > 0$ we have

$$u_\varepsilon(t) = \int_{-\infty}^t S_\varepsilon(t-s) f\left(\frac{\cdot}{\varepsilon}, u_\varepsilon(s, \cdot)\right) ds + \int_{-\infty}^t S_\varepsilon(t-s) \sigma\left(\frac{\cdot}{\varepsilon}, u_\varepsilon(s, \cdot)\right) dW(s)$$

4 and

$$u_0(t) = \int_{-\infty}^t S_0(t-s) f_0(u_0(s, \cdot)) ds + \int_{-\infty}^t S_0(t-s) \sigma_0(u_0(s, \cdot)) dW(s).$$

5 For $n \geq 1$ denote

$$u_\varepsilon^n(t) := \int_{-n}^t S_\varepsilon(t-s) f\left(\frac{\cdot}{\varepsilon}, u_\varepsilon(s, \cdot)\right) ds + \int_{-n}^t S_\varepsilon(t-s) \sigma\left(\frac{\cdot}{\varepsilon}, u_\varepsilon(s, \cdot)\right) dW(s)$$

6 and

$$u_0^n(t) := \int_{-n}^t S_0(t-s) f_0(u_0(s, \cdot)) ds + \int_{-n}^t S_0(t-s) \sigma_0(u_0(s, \cdot)) dW(s).$$

7 Then

$$\begin{aligned} \mathbb{E} \|u_\varepsilon^n(t) - u_\varepsilon(t)\|_\rho^2 &\leq 2\mathbb{E} \left\| \int_{-\infty}^{-n} S_\varepsilon(t-s) f\left(\frac{\cdot}{\varepsilon}, u_\varepsilon(s, \cdot)\right) ds \right\|_\rho^2 \\ &\quad + 2\mathbb{E} \left\| \int_{-\infty}^{-n} S_\varepsilon(t-s) \sigma\left(\frac{\cdot}{\varepsilon}, u_\varepsilon(s, \cdot)\right) dW(s) \right\|_\rho^2 \\ &\leq B_1 e^{-2\gamma(t+n)} \rightarrow 0, \quad n \rightarrow \infty, \end{aligned} \quad (4.9)$$

8 where B_1 is independent of n, t and ε . Similarly,

$$\mathbb{E} \|u_0^n(t) - u_0(t)\|_\rho^2 \leq B_1 e^{-2\gamma(t+n)} \rightarrow 0, \quad n \rightarrow \infty.$$

9 Then

$$\|u_\varepsilon(t) - u_0(t)\|_\rho^2 \leq 3(\|u_\varepsilon(t) - u_\varepsilon^n(t)\|_\rho^2 + \|u_0(t) - u_0^n(t)\|_\rho^2 + \|u_0^n(t) - u_\varepsilon^n(t)\|_\rho^2). \quad (4.10)$$

10 We now estimate the last term in (4.10).

$$\begin{aligned} &\mathbb{E} \|u_0^n(t) - u_\varepsilon^n(t)\|_\rho^2 \\ &\leq 2\mathbb{E} \left\| \int_{-n}^t \left(S_\varepsilon(t-s) f\left(\frac{\cdot}{\varepsilon}, u_\varepsilon(s, \cdot)\right) - S_0(t-s) f_0(u_0(s, \cdot)) \right) ds \right\|_\rho^2 \\ &\quad + 2\mathbb{E} \left\| \int_{-n}^t \left(S_\varepsilon(t-s) \sigma\left(\frac{\cdot}{\varepsilon}, u_\varepsilon(s, \cdot)\right) - S_0(t-s) \sigma_0(u_0(s, \cdot)) \right) dW(s) \right\|_\rho^2 \\ &:= J_1 + J_2. \end{aligned} \quad (4.11)$$

O. Misiats, O. Stanzhytskyi & N. K. Yip

1 Once we add and subtract $S_\varepsilon(t-s)f(\cdot/\varepsilon, u_\varepsilon^n)$ and $S_0(t-s)f(u_0)$, the term J_1 may
2 be estimated as

$$J_1 \leq 3(J_{11} + J_{12} + J_{13})$$

3 Using (4.9) and the Lipschitz property, one may obtain the bounds for J_{11}

$$J_{11} := C_1 L^2 \mathbb{E} \left(\int_{-n}^t e^{-\gamma(t-s)} \|u_\varepsilon(s) - u_\varepsilon^n(s)\|_\rho ds \right)^2 \leq B_2 e^{-\gamma(t+n)}$$

4 and

$$J_{12} := C_1 L^2 \mathbb{E} \left(\int_{-n}^t e^{-\gamma(t-s)} \|u_0(s) - u_0^n(s)\|_\rho ds \right)^2 \leq B_2 e^{-\gamma(t+n)}.$$

5 We now proceed with the estimate for J_{13} . Using (4.5), we have

$$\begin{aligned} J_{13} &\leq 2 \mathbb{E} \left(\int_{-n}^t \sqrt{C_1} e^{-\gamma(t-s)} L \|u_\varepsilon^n(s) - u_0^n(s)\|_\rho ds \right)^2 \\ &\quad + 2 \mathbb{E} \left(\int_{-n}^t \left\| S_\varepsilon(t-s)f\left(\frac{\cdot}{\varepsilon}, u_0^n(s)\right) - S_0(t-s)f_0(u_0^n(s)) \right\|_\rho ds \right)^2. \end{aligned}$$

6 The last term may be rewritten as

$$\begin{aligned} &\mathbb{E} \left(\int_{-n}^t \left\| S_\varepsilon(t-s)f\left(\frac{\cdot}{\varepsilon}, u_0^n(s)\right) - S_0(t-s)f_0(u_0^n(s)) \right\|_\rho ds \right)^2 \\ &\leq \int_{-n}^t e^{-(t-s)} \varphi_1(t, s, \varepsilon, n) ds, \end{aligned}$$

7 where

$$\begin{aligned} \varphi_1(t, s, \varepsilon, n) &:= \mathbb{E} \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} K_\varepsilon(x, y, t-s) f\left(\frac{y}{\varepsilon}, u_0^n(s)\right) \right. \\ &\quad \left. - K_0(x, y, t-s) f_0(u_0^n(s)) \right)^2 \rho(x) dx. \end{aligned} \quad (4.12)$$

8 **Lemma 4.5.** *For fixed t, s and n , $\varphi_1(t, s, n, \varepsilon)$ converges to 0 as $\varepsilon \rightarrow 0$.*

9 **Proof.** For $R > 0$ we can write (4.12) as

$$\varphi_1(t, s, n, \varepsilon) = \int_{|x| \geq R} \{\dots\} \rho(x) dx + \int_{|x| < R} \{\dots\} \rho(x) dx := J_{131} + J_{132}.$$

10 Next, we are going to use the facts that $|f(x, u)| \leq B_{f, \sigma}$ is uniformly bounded, and
11 that $\int_{\mathbb{R}^d} K_\varepsilon(x, y, t) dy = \int_{\mathbb{R}^d} K_0(x, y, t) dy = 1$, in order to estimate J_{131} . We have

$$J_{131} \leq 4B_{f, \sigma}^2 \int_{|x| \geq R} \rho(x) dx.$$

12 Therefore, for any $\eta > 0$ there exists $R > 0$ such that

$$J_{131} \leq \frac{\eta}{2}.$$

1 For such R we proceed with the estimate for

$$J_{132} := \mathbb{E} \int_{|x| < R} \left(\int_{\mathbb{R}^d} (K_\varepsilon(x, y, t-s) f\left(\frac{y}{\varepsilon}, u_0^n(s)\right) - K_0(x, y, t-s) f_0(u_0^n(s))) dy \right)^2 \rho(x) dx$$

2 Let $h(x) \in L^2(\Omega)$ be a fixed function. By Lemmas 4.1 and 4.2

$$\psi(x, y) := f(y, u_0^n(x, s, \omega)) h(x) \in L^1[B_R(0), C_\#(Y)].$$

3 thus for a.e. ω

$$\int_{|x| < R} \psi\left(x, \frac{x}{\varepsilon}\right) dx \rightarrow \int_{\Pi} \int_{|x| < R} h(x) f(y, u_0^n(s, x, \omega)) dx dy, \quad \varepsilon \rightarrow 0.$$

4 This, in turn, implies that

$$f\left(\frac{x}{\varepsilon}, u_0^n(s, x, \omega)\right) \rightharpoonup f_0(u_0^n(s, x, \omega)) \quad \text{in } L^2_{\text{loc}}(\mathbb{R}^d), \quad \varepsilon \rightarrow 0.$$

5 We are now in position to apply the convergence result [43, Chap. 2, Theorem 1]
6 to conclude that

$$\int_{\mathbb{R}^d} \left(K_\varepsilon(x, y, t-s) f\left(\frac{y}{\varepsilon}, u_0^n(s)\right) - K_0(x, y, t-s) f_0(u_0^n(s)) \right) dy \rightarrow 0, \quad \varepsilon \rightarrow 0$$

7 in $L^2_{\text{loc}}(\mathbb{R}^d)$. By Dominated Convergence Theorem, choosing $\varepsilon > 0$ small enough,
8 we have

$$J_{132} < \frac{\eta}{2},$$

9 or $\varphi_1(t, s, n, \varepsilon) < \eta$, which concludes the proof of Lemma 4.5. $\square$

10 Therefore,

$$J_{13} \leq \frac{2C_\nu^2}{\gamma} L^2 \int_{-n}^t e^{-\gamma(t-s)} \mathbb{E} \|u_\varepsilon^n(s) - u_0^n(s)\|_\rho^2 ds + \int_{-n}^t e^{-(t-s)} \varphi_1(t, s, \varepsilon, n) ds,$$

11 and combining the bounds for J_{11} , J_{12} and J_{13} we arrive at

$$\begin{aligned} J_1 &\leq B_1 e^{-\gamma(t+n)} + \frac{6C_\nu^2 L^2}{\gamma} \int_{-n}^t e^{-\gamma(t-s)} \mathbb{E} \|u_\varepsilon^n(s) - u_0^n(s)\|_\rho^2 ds \\ &\quad + 3 \int_{-n}^t e^{-(t-s)} \varphi_1(t, s, \varepsilon, n) ds. \end{aligned} \quad (4.13)$$

12 We now estimate the stochastic contribution J_2 . In a similar manner, we write

$$J_2 \leq 3(J_{21} + J_{22} + J_{23}),$$

O. Misiats, O. Stanzhytskyi & N. K. Yip

1 where

$$\begin{aligned} J_{21} &:= \mathbb{E} \left\| \int_{-n}^t S_\varepsilon(t-s)(\sigma(\cdot/\varepsilon, u_\varepsilon) - \sigma(\cdot/\varepsilon, u_\varepsilon^n)) dW(s) \right\|_\rho^2 \\ &\leq C_\nu a \int_{-n}^t e^{-2\gamma(t-s)} L^2 \mathbb{E} \|u_\varepsilon(s) - u_\varepsilon^n(s)\|_\rho^2 ds \leq B_3 e^{-\gamma(t+n)}. \end{aligned}$$

2 Analogously,

$$J_{22} := \mathbb{E} \left\| \int_{-n}^t S_0(t-s)(\sigma_0(u_0) - \sigma_0(u_0^n)) dW(s) \right\|_\rho^2 \leq B_3 e^{-\gamma(t+n)}.$$

3 We proceed with the estimate for J_{23} :

$$\begin{aligned} J_{23} &= \mathbb{E} \left\| \int_{-n}^t S_\varepsilon(t-s)(\sigma(\cdot/\varepsilon, u_\varepsilon^n) - \sigma_0(u_0^n)) dW(s) \right\|_\rho^2 \\ &= \sum_{k=1}^{\infty} a_k \int_{-n}^t \mathbb{E} \|S_\varepsilon(t-s)\sigma(\cdot/\varepsilon, u_\varepsilon^n)e_k(\cdot) - S_0(t-s)\sigma_0(u_0^n)e_k(\cdot)\|_\rho^2 ds. \end{aligned}$$

4 Adding and subtracting $S_\varepsilon(t-s)\sigma(\cdot/\varepsilon, u_0^n)$, we have

$$\begin{aligned} J_{23} &\leq 2 \sum_{k=1}^{\infty} a_k \int_{-n}^t \mathbb{E} \|S_\varepsilon(t-s)[\sigma(\cdot/\varepsilon, u_\varepsilon^n) - \sigma(\cdot/\varepsilon, u_0^n)]e_k(\cdot)\|_\rho^2 ds \\ &\quad + 2 \sum_{k=1}^{\infty} a_k \int_{-n}^t \mathbb{E} \|S_\varepsilon(t-s)\sigma(\cdot/\varepsilon, u_0^n)e_k(\cdot) - S_0(t-s)\sigma_0(u_0^n)e_k(\cdot)\|_\rho^2 ds \\ &\leq 2aC_\nu^2 L^2 \int_{-n}^t e^{-\gamma(t-s)} \mathbb{E} \|u_\varepsilon^n(s) - u_0^n(s)\|_\rho^2 ds \\ &\quad + 2 \sum_{k=1}^{\infty} a_k \int_{-n}^t e^{-\gamma(t-s)} \varphi_2(t, s, \varepsilon, n, k) ds, \end{aligned}$$

5 where

$$\begin{aligned} \varphi_2(t, s, \varepsilon, n, k) &:= \mathbb{E} \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} [K_\varepsilon(x, y, t-s)\sigma(y/\varepsilon, u_0^n)e_k(y) \right. \\ &\quad \left. - K_0(x, y, t-s)\sigma_0(u_0^n)e_k(y)] dy \right)^2 \rho(x) dx. \end{aligned}$$

6 By an argument, identical to Lemma 4.5, $\varphi_2(t, s, \varepsilon, n, k)$ is uniformly bounded by
7 an absolute constant, and $\varphi_2(t, s, \varepsilon, n, k) \rightarrow 0, \varepsilon \rightarrow 0$. Consequently,

$$\varphi_3(t, s, \varepsilon, n) := \sum_{k=1}^{\infty} a_k \varphi_2(t, s, \varepsilon, n, k) \rightarrow 0, \quad \varepsilon \rightarrow 0.$$

1 Altogether, we have

$$J_2 \leq B_3 e^{-\gamma(t+n)} + 6aC_\nu^2 L^2 \int_{-n}^t e^{-\gamma(t-s)} \mathbb{E} \|u_\varepsilon^n(s) - u_0^n(s)\|_\rho^2 ds \\ + 6 \int_{-n}^t e^{-(t-s)} \varphi_3(t, s, \varepsilon, n) ds]. \quad (4.14)$$

2 Now, combining (4.14) with (4.13) and (4.11), we have

$$\mathbb{E} \|u_\varepsilon^n(t) - u_0^n(t)\|_\rho^2 \leq B_4 e^{-\gamma(t+n)} + \int_{-n}^t e^{-\gamma(t-s)} \varphi_4(t, s, \varepsilon, n) ds \quad (4.15)$$

$$+ 12 \left(\frac{1}{\gamma} + a \right) C_\nu^2 L^2 \int_{-n}^t e^{-\gamma(t-s)} \mathbb{E} \|u_\varepsilon^n(s) - u_0^n(s)\|_\rho^2 ds, \quad (4.16)$$

3 where $\varphi_4(t, \varepsilon, n) = 12 \int_{-n}^t (\varphi_1 + \varphi_3)(t, s, \varepsilon, n) ds \rightarrow 0, \varepsilon \rightarrow 0$. Denoting

$$z(t) := e^{\gamma t} \mathbb{E} \|u_\varepsilon^n(t) - u_0^n(t)\|_\rho^2,$$

4 and

$$H := 12 \left(\frac{1}{\gamma} + a \right) C_\nu L^2$$

5 (4.16) may be rewritten as

$$z(t) \leq B_4 e^{-\gamma n} + \varphi_4(t, \varepsilon, n) + H \int_{-n}^t z(s) ds,$$

6 and hence, by Gronwall inequality

$$z(t) \leq B_4 e^{-\gamma n} + \varphi_4(t, \varepsilon, n) + \int_{-n}^t e^{H(t-s)} (B_4 e^{-\gamma n} + \varphi_4(s, \varepsilon, n)) ds.$$

7 If (4.8) holds, for fixed $t \geq 0$ and any $\eta > 0$ we can choose $n \geq 1$ such that

$$B_4 e^{-\gamma n} + B_4 e^{-\gamma n} \int_{-n}^t e^{H(t-s)} ds < \frac{\eta}{2}.$$

8 For such n we may now choose $\varepsilon > 0$ such that

$$\varphi_4(t, \varepsilon, n) + \int_{-n}^t e^{H(t-s)} \hat{\varphi}(s, \varepsilon, n) ds < \frac{\eta}{2}$$

9 and hence

$$z(t) = e^{\gamma t} \mathbb{E} \|u_\varepsilon^n(t) - u_0^n(t)\|_\rho^2 < \eta.$$

10 This completes the proof of Theorem 1.5.

11 **Proof of Theorem 1.6.** Let $S_\varepsilon(t)$ and $S_0(t)$ be the semigroups generated by A_ε
12 and A_0 with Dirichlet boundary conditions $u = 0$ on ∂G . Then

$$\|S_\varepsilon \varphi\|_{L^2(G)} \leq \tilde{C}_\nu e^{-\tilde{\gamma} t} \|\varphi\|_{L^2(G)}$$

O. Misiats, O. Stanzhytskyi & N. K. Yip

1 and

$$\|S_0\varphi\|_{L^2(G)} \leq \tilde{C}_\nu e^{-\tilde{\gamma}t} \|\varphi\|_{L^2(G)},$$

2 where $\tilde{\gamma} > 0$ depends on ν and on the first eigenvalue of the Laplace operator in G .
 3 Provided (4.6)–(4.8) hold with C_ν and γ replaced with $\tilde{C}_\nu$ and $\tilde{\gamma}$ correspondingly,
 4 the rest of the proof of Theorem 1.6 follows the lines of Theorem 1.5. $\square$

5 Acknowledgments

6 This work was partially completed while Oleksandr Misiats was a Golomb Visiting
 7 Assistant Professor at Purdue University. The author is thankful to the Department
 8 of Mathematics at Purdue University for the support and hospitality. The research
 9 of Oleksandr Misiats was partially supported by the grant DMS-0967140.

10 References

- 11 1. G. Allaire, Homogenization and two-scale convergence, *SIAM J. Math. Anal.* **6** (1991)
 12 1482–1518.
- 13 2. S. Assing and R. Mantey, Invariant measures for stochastic heat equations with
 14 unbounded coefficients, *Stochastic Process. Appl.* **103** (2003) 237–256.
- 15 3. A. Bensoussan, Homogenization of a class of stochastic partial differential equations,
 16 *Prog. Nonlinear Differential Equations Appl.* **5** (1991) 47–65.
- 17 4. H. Bessaih, Y. Efendiev and F. Maris, Homogenization of Brinkman flows in heteroge-
 18 neous dynamic media, *Stoch. Partial Differential Equations: Anal. Comput.* **3** (2015)
 19 479–505.
- 20 5. V. Bogachev and M. Rockner, Elliptic equations for measures on infinite-dimensional
 21 spaces and applications, *Probab. Theory Related Fields* **120** (2001) 445–496.
- 22 6. Z. Brzezniak and D. Gatarek, Martingale solutions and invariant measures for stochas-
 23 tic evolution equations in Banach spaces, *Stochastic Process. Appl.* **84** (1999) 87–225.
- 24 7. V. Capasso and D. Fortunato, Stability results for semilinear evolution equations
 25 and their applications to some reaction–diffusion problems, *SIAM J. Appl. Math.* **39**
 26 (1980) 34–47.
- 27 8. S. Cerrai, *Second Order PDE's in Finite and Infinite Dimension* (Springer, 2001).
- 28 9. S. Cerrai, Stochastic reaction–diffusion systems with multiplicative noise and non-
 29 Lipschitz reaction term, *Probab. Theory Related Fields* **125** (2003) 271–304.
- 30 10. S. Cerrai, Asymptotic behavior of systems of stochastic partial differential equations
 31 with multiplicative noise, in *Stochastic Partial Differential Equations and Applications*
 32 *VII*, Lecture Notes in Pure and Applied Mathematics, Vol. 245 (CRC Press, 2006),
 33 pp. 61–75.
- 34 11. P.-L. Choi, *Stochastic Partial Differential Equations* (Chapman & Hall/CRC, 2007).
- 35 12. G. Da Prato, D. Gatarek and J. Zabczyk, Invariant measures for semilinear stochastic
 36 equations, *Stoch. Anal. Appl.* **10** (1992) 387–408.
- 37 13. G. Da Prato and J. Zabczyk, *Stochastic Equations in Infinite Dimensions* (Cambridge
 38 University Press, 1992).
- 39 14. G. Da Prato and J. Zabczyk, Non-explosion, boundedness, and ergodicity for stochas-
 40 tic semilinear equations, *J. Differential Equations* **98** (1992) 181–195.
- 41 15. G. Da Prato and J. Zabczyk, *Ergodicity for Infinite-Dimensional Systems* (Cambridge
 42 University Press, 1996).

- 1 16. D. Dawson, Stochastic evolution equations, *Mathematical Biosciences* **154** (1972) 187–
- 2 316.
- 3 17. N. Dirr and N. K. Yip, Pinning and de-pinning phenomena in front propagation in
- 4 heterogeneous media, *Interfaces Free Bound.* **8** (2006) 79–109.
- 5 18. Y. Du and S. Hsu, On a nonlocal reaction–diffusion problem arising from the modeling
- 6 of phytoplankton growth, *SIAM J. Math. Anal.* **42** (2010) 1305–1333.
- 7 19. J. Eckmann and M. Hairer, Invariant measures for stochastic partial differential equa-
- 8 tions in unbounded domains, *Nonlinearity* **14** (2001) 133–151.
- 9 20. I. Gikhman and A. Skorokhod, *Stochastic Differential Equations* (Springer, 1972).
- 10 21. B. Goldys and B. Maslowski, Uniform exponential ergodicity of stochastic dissipative
- 11 systems *Czechoslovak Math. J.* **51** (2001) 745–762.
- 12 22. D. Henry, *Geometric Theory of Semilinear Parabolic Equations*, Lecture Notes in
- 13 Mathematics (1981).
- 14 23. N. Ichihara, Homogenization for stochastic partial differential equations derived from
- 15 nonlinear filtering with feedback, *J. Math. Soc. Japan* **57** (2005) 593–603.
- 16 24. N. Kryloff and N. Bogoliouboff, La théorie générale de la mesure dans son application
- 17 à l'étude des systèmes dynamiques de la mécanique non linéaire, *Ann. of Math.* (2)
- 18 **38** (1937) 65–113.
- 19 25. N. V. Krylov, A relatively short proof of Itos formula for SPDEs and its applications,
- 20 *Stoch. Partial Differential Equations: Anal. Comput.* **1** (2013) 152–174.
- 21 26. R. Manthey, The long-time behaviour of the solutions to semilinear stochastic partial
- 22 differential equations on the whole space, *Math. Bohem.* **126** (2001) 15–39.
- 23 27. R. Manthey and T. Zausinger, Stochastic evolution equations in L^2_ρ , *Stoch. Stoch.*
- 24 *Rep.* **66** (1999) 37–85.
- 25 28. B. Maslowski and J. Seidler, On sequentially weakly Feller solutions to SPDEs, *Atti*
- 26 *Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. Rend. Lincei Mat. Appl.* **10** (1999) 69–78.
- 27 29. J. Mattingly and E. Pardoux, Invariant measure selection by noise, an example, *Dis-*
- 28 *crete Contin. Dynam. Syst. A* **34** (2014) 237–256.
- 29 30. H. McKean, Nagumo's equation, *Adv. Math.* **4** (1970) 209–223.
- 30 31. O. Misiats, O. Stanzhytskyi and N. K. Yip, Existence and uniqueness of invariant
- 31 measures for stochastic reaction–diffusion equations in unbounded domains, *J. Theor.*
- 32 *Probab.* **29** (2016) 996–1026.
- 33 32. O. Misiats, O. Stanzhytskyi and N. K. Yip, Ito's formula in infinite dimension and
- 34 its application to the existence of invariant measures, submitted.
- 35 33. C. Mueller, Coupling and invariant measures for the heat equation with noise, *Ann.*
- 36 *Probab.* **21** (1993) 2189–2199.
- 37 34. C. Prévôt and M. Röckner, *A Concise Course on Stochastic Partial Differential Equa-*
- 38 *tions*, Lecture Notes in Mathematics (Springer, Berlin, 2007).
- 39 35. Y. Prokhorov and A. Shiryaev, *Probability Theory III: Stochastic Calculus* (Springer,
- 40 1997).
- 41 36. A. Samoilenko, M. Perestyuk and I. Parasyuk, *Differential Equations* (2012).
- 42 37. M. Sanz-Solé and M. Sarrà, Hölder continuity for the stochastic heat equation with
- 43 spatially correlated noise, in *Seminar on Stochastic Analysis, Random Fields and*
- 44 *Applications, III*, Vol. 52 (2002), pp. 259–268.
- 45 38. G. Tessitore and J. Zabczyk, Invariant measures for stochastic heat equations, *Probab.*
- 46 *Math. Statist.* **2** (1998) 271–287.
- 47 39. E. Titchmarsh, *Eigenfunction Expansions Associated With Second-Order Differential*
- 48 *Equations, Part 2* (Clarendon Press, 1958).
- 49 40. V. Volpert, *Elliptic Partial Differential Equations II: Reaction–Diffusion Equations*
- 50 (Springer, 2014).

AQ: Please provide
publisher details.

AQ: Please update.

AQ: Please provide
publisher details.

O. Misiats, O. Stanzhytskyi & N. K. Yip

- 1 41. V. Volpert and V. Vougalter, Existence of stationary pulses for nonlocal reaction–
2 diffusion equations, *Documenta Math.* **19** (2014) 1141–1153.
- 3 42. W. Wang, D. Cao and J. Duan, Effective macroscopic dynamics of stochastic partial
4 differential equations in perforated domains, *SIAM J. Math. Anal.* **38** (2007) 1508–
5 1527.
- 6 43. V. Zhikov, S. Kozlov and O. Oleinik, *Homogenization of Differential Operators and*
7 *Integral Functionals* (Springer, 1994).

AQ: Please provide the
citation of Refs. 11, 25,
34, 16, 20, 17, 26, 27,
30, 36, 37, 12, 41, 18,
7, 35, 22, 14.